Figure 3. Enlarged rings turn more slowly and respond less to weak boundary input

Enlargement reduces boundary susceptibility, with a substantial cytosolic-conduction penalty in Wave-pinning and smaller native penalties in Holmes and Spring. Goryachev loses gain much more steeply than the conduction law predicts.

A, Native Wave-pinning C30 active-concentration kymographs for a cue overlapping the incumbent's 10%-contrast contour (edge gap −1) or separated from it (gap 4). Time starts at cue onset; cyan bars show spatial cue support. White dashed lines mark withdrawal after 95% of the target heading change; subsequent single-domain relaxation is included. Native cue peak 0.1 and support 3 are unchanged.

B, Time to turn halfway toward a continuous cue against its edge gap, at C30, 37.8 and 47.6 (Spring resting sizes 2.5, 3.15 and 3.97). Shades denote size within each model. Open triangles with upward arrows are right-censored lower bounds; grey spans mark gaps at which heading was undefined at one or more sizes. These strong-cue turning times are distinct from weak-pulse susceptibility.

C, Susceptibility to a weak Gaussian pulse, permanent displacement per peak×duration dose, against position at C40 (Spring resting size 2). Colored lines use the native translation-mode adjoint at f=0.001 (Spring mean input 0.004); grey curves and right axes show the same stationary mean active profile. The Gaussian SD is 1.5 (Spring 0.15). Signed gain is localized to the two lateral boundaries.

D, Wave-pinning shared permanent boundary displacement against edge-pulse peak divided by reference input 0.1, at C20 and C60; duration 50. Dashed lines use the weak-input gain. Ten-percent departures are bracketed between 5–20% of reference input in Wave-pinning, 2.6–5% in Holmes, 1.3–2.6% in Spring and 2.6–20% in Goryachev (neighboring tested amplitudes, pooled size ranges). Dose-matched duration sweeps agree within 0.2%.

E, Native edge gain normalized to each model's smallest size, against relative resting size. Lines show normalized $G=(\lambda/2)/(1+C/C^*)$. Wave-pinning $C^*=76.28$ comes from the independently solved isolated front, $\kappa=1.28149$, $\Delta u=0.96085$, and the stationary native domain fraction θ; no ring gain is fitted. Holmes $C^*=403.45$ comes from the native-D inverse-gain regression $1/G=a+bC$, $C^*=a/b$, corroborated by the diffusivity sweep. Spring transport $C^*=49.00$ comes from a through-origin free-conduction penalty regression $G(1000,C)/G(D,C)-1$ against C/D over $D\geq 1$. Its dashed curve is the pure conduction law; its solid curve additionally includes the independently measured $G(1000,C)/G(1000,C_{min})$ rise. The apparent native inverse-gain crossover is 142.30, which reflects cancellation by this intrinsic rise and is not equated with the transport crossover. Holmes's alternative penalty fit gives 469.06. Goryachev has no conduction-law curve. Spring's actual circumference includes its small global extension. The E native points retain round6 mean input 0.004; the Spring diffusivity sweep uses pilot mean 0.0004.

F, Wave-pinning adjoint edge gain against circumference at inactive diffusivities 2.5, 5, 10 (native), 20, 40, 100 and 1000. Every point uses the same weak mean input and unchanged reaction kinetics. All 35 Wave-pinning preparations retain a polarized branch, including $D_v=2.5$ at C60 where the pilot reported domain loss. Fast cytosol largely removes the native size dependence. All quantities use model-native units; none is a calibrated migration velocity or

Figure 3. Enlarged rings turn more slowly and respond less to weak boundary input · continued

biochemical dose.

$$G = \frac{\lambda/2}{1 + C/C^*}, \qquad C^* = \frac{2D_v}{\kappa \Delta u \theta (1-\theta)} \tag{3.1}$$

Wave-pinning crossover uses the isolated front and stationary domain fraction; Holmes uses the native inverse-gain regression and Spring uses the diffusivity-penalty regression, with intrinsic free-conduction rise shown explicitly.